Strong Method Problem Solving

(Topic 7-part b)



Course Contents

Again..Selected topics for our course. Covering all of AI is impossible!

Key topics include:

- Introduction to Artificial Intelligence (AI)
- Knowledge Representation and Search
- Introduction to AI Programming
- Problem Solving Using Search
- Exhaustive Search Algorithm
- Heuristic Search
- Techniques and Mechanisms of Search Algorithm
- Knowledge Representation Issues and Concepts
- Strong Method Problem Solving
- Soft Computing and Machine Learning

Strong Method Problem Solving (knowledge-intensive or strong method)

- Introduction to expert system
- Rule-based expert system
- Case-based
- Model-based
- Hybrid expert system

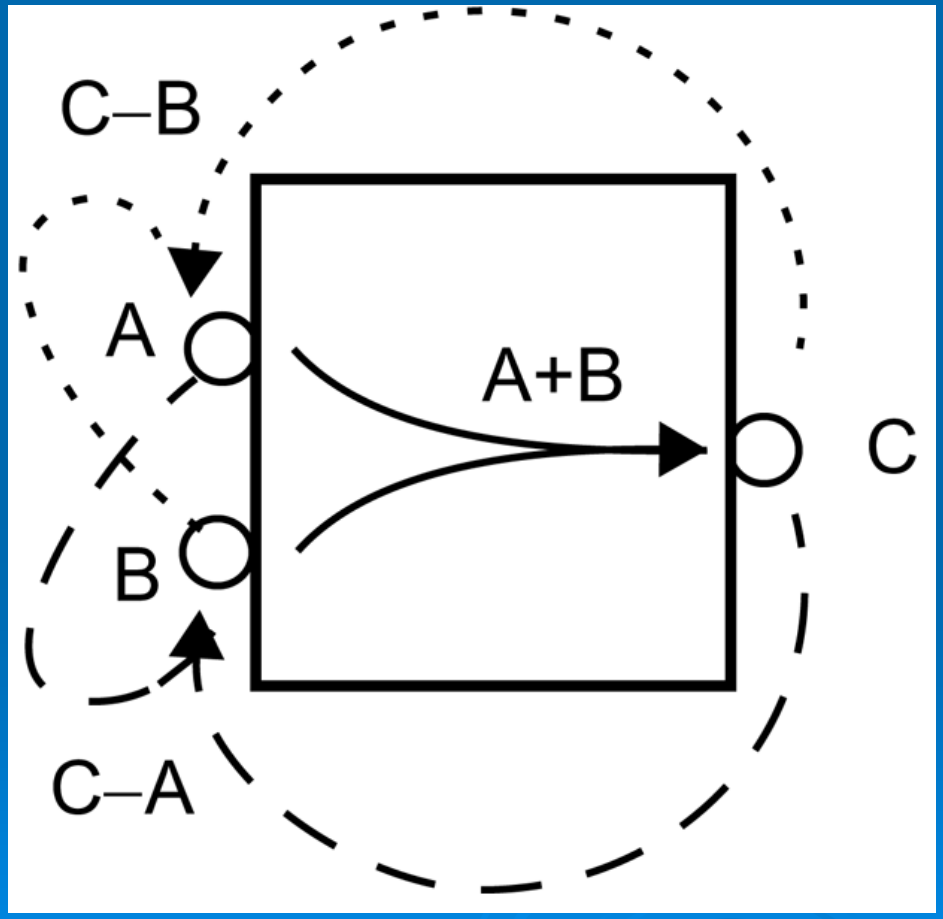
Model-based reasoning

- Often expert systems applied heuristics in inappropriate situations
- Limitation *model-based* tries to address
- What is model-based system?
  - A knowledge-based reasoner whose analysis is founded directly on the specifications and functionality of a physical system
  - Creates a software simulation
  - i.e create model for electronic circuit

What is model-based (MB) reasoning system?

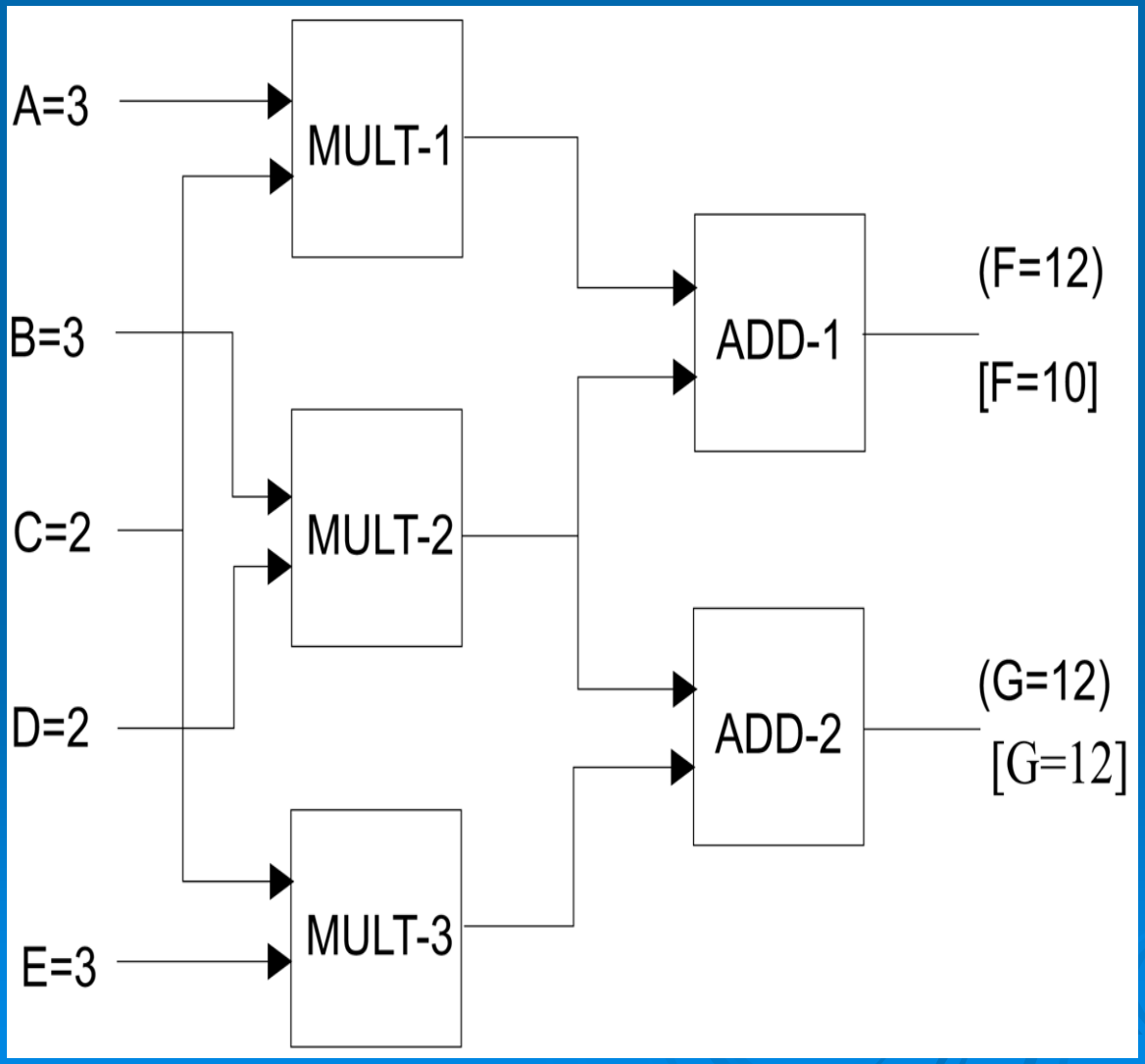
- i.e. in trouble shooting faults in physical system,
  - Model leads to predicted behaviors
  - Fault ~ discrepancies between predicted and expected behavior
  - MB tells user what to expect, when observations differ from expectations, discrepancies leads to identification of faults

Model-Based reasoner: Example 1
The behavior description of an adder, after Davis and Hamscher (1988).



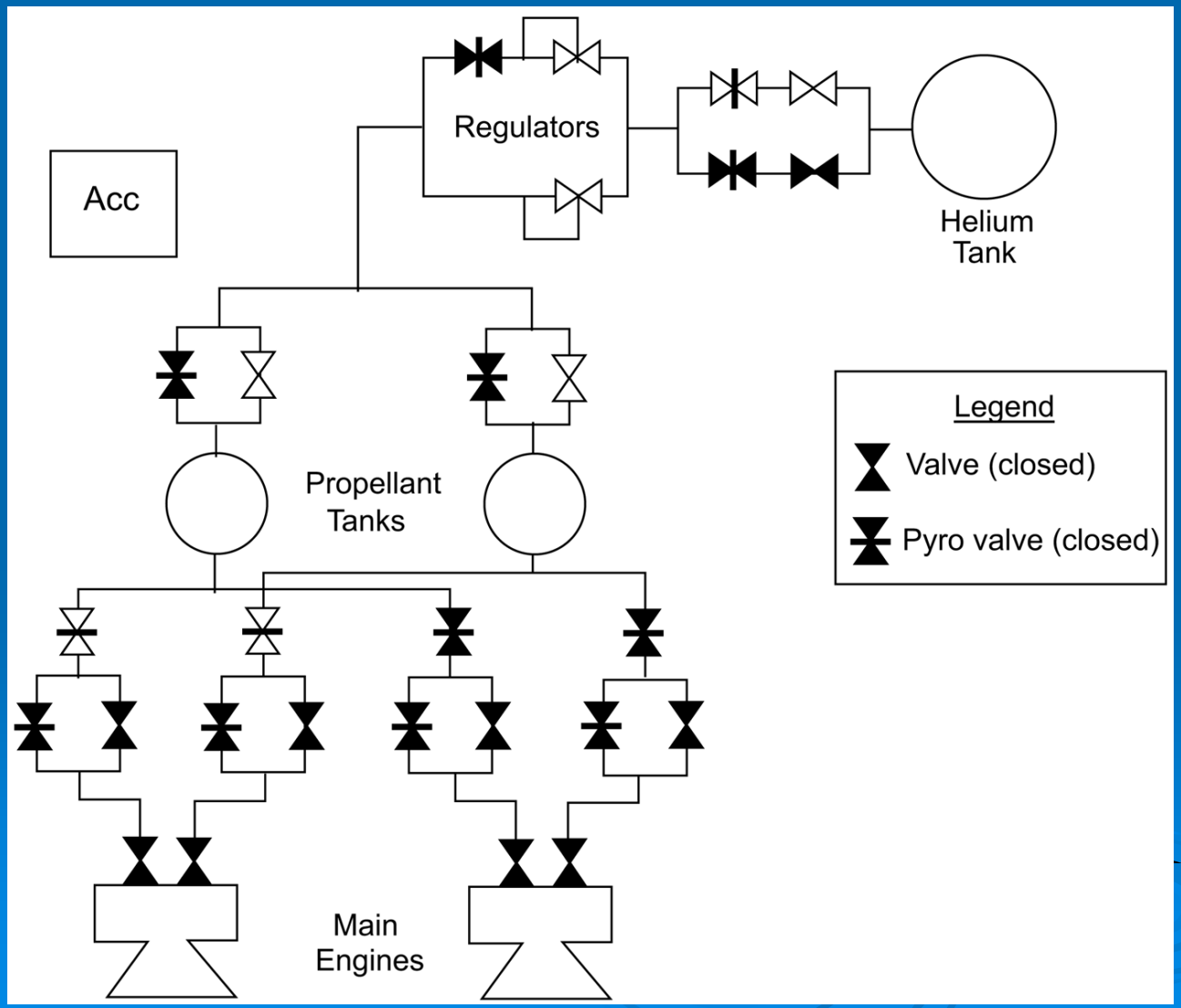
• device and circuit analysis – the adder

Model-Based reasoner: Example 2
Taking advantage of direction of information flow, after Davis and Hamscher (1988).



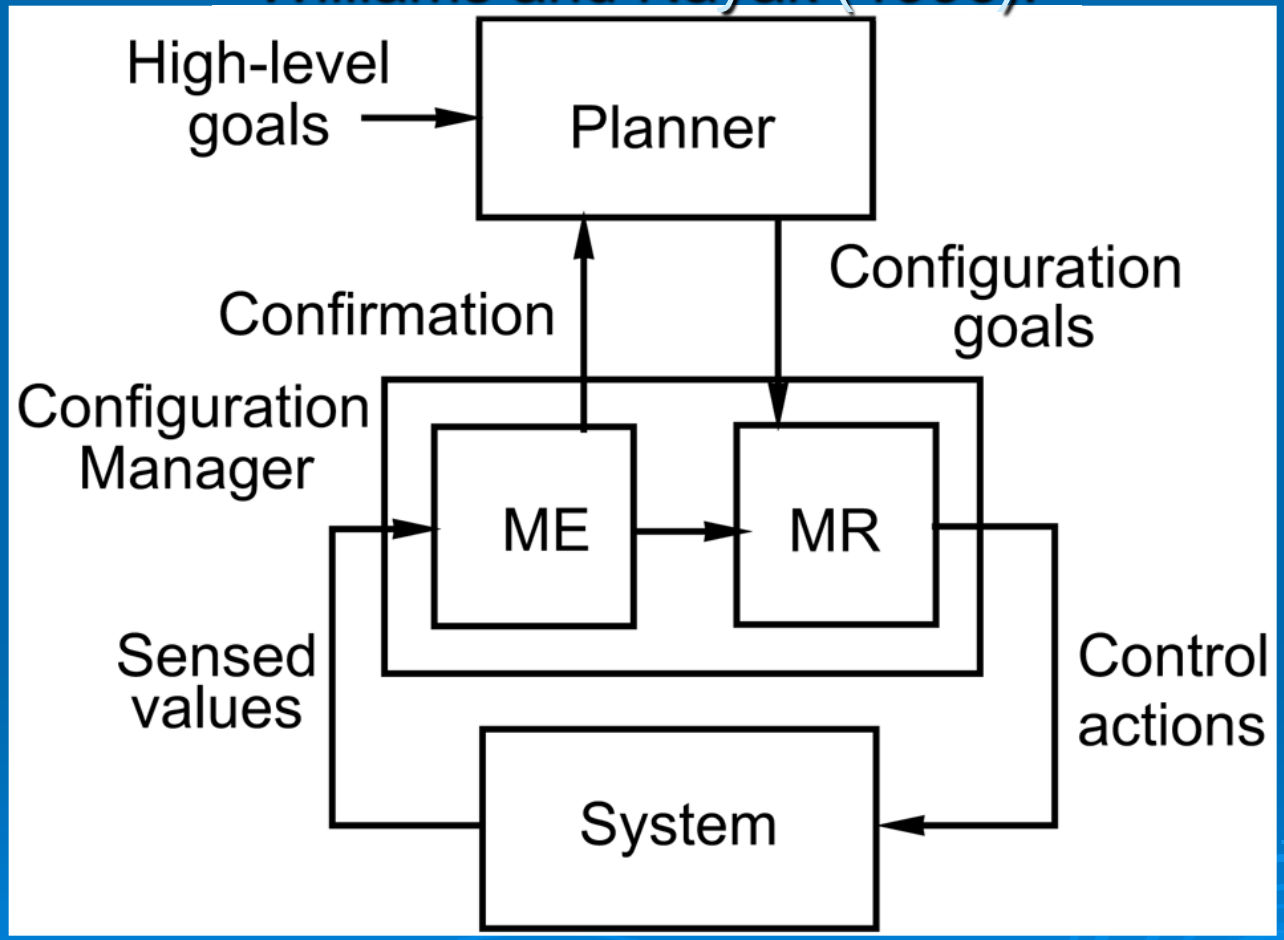
- circuit of three multipliers and two adders
- task is to determine where the fault lies that will explain discrepancies

Model-Based reasoner: Example 3
A schematic of the simplified Livingstone propulsion system, from Williams and Nayak (1996)-NASA.



- idealized schematic of main engine subassembly-complex spacecraft
- its function to isolate failed components permanently

Model-Based reasoner: Example 4
A model-based configuration management system, from Williams and Nayak (1996).



• model-based configuration system uses a model to determine desired control sequence for estimation and reconfiguration stage

What is case-based (CBR) reasoning system?

- Knowledge-intensive technique that supports the reuse of past experience in a problem domain to address new situations
- Another powerful strategy experts use – reasoning from case, examples of past problems and solution
- CBR uses explicit database of problem solutions to address new problem solving situations

What is case-based (CBR) reasoning system?

- Solutions may be collected from:
  - Human experts
  - Results of previous search-based success or failures
  - Medical education – depends heavily on case histories and experience with other patients
  - Lawyers- select past law cases to convince court
  - Programmers- reuse code

What is case-based reasoning (CBR) system?

Case-based reasoners share a common structure. For each new problem they:

1. Retrieve appropriate cases from memory-use heuristic to choose similar cases. Eg. Two patients have same symptoms they might have same disease hence same treatment. For efficient retrieval, cases are indexed

2. Modify a retrieved case so that it will apply to the current situation-analytic method, eg. To determine appropriate temperatures or materials for welding

3. Apply the transformed case

4. Save the solution, with a record of success or failure, for future use-updating the index structure, may use clustering algorithm and machine learning

CBR ….

Kolodner (1993) offers a set of possible preference heuristics to help organize the storage and retrieval of cases. These include:

1. *Goal-directed preference.* Organize cases, at least in part, by goal descriptions. Retrieve cases that have the same goal as the current situation.

2. *Salient-feature preference.* Prefer cases that match the most important features or those matching the largest number of important features.

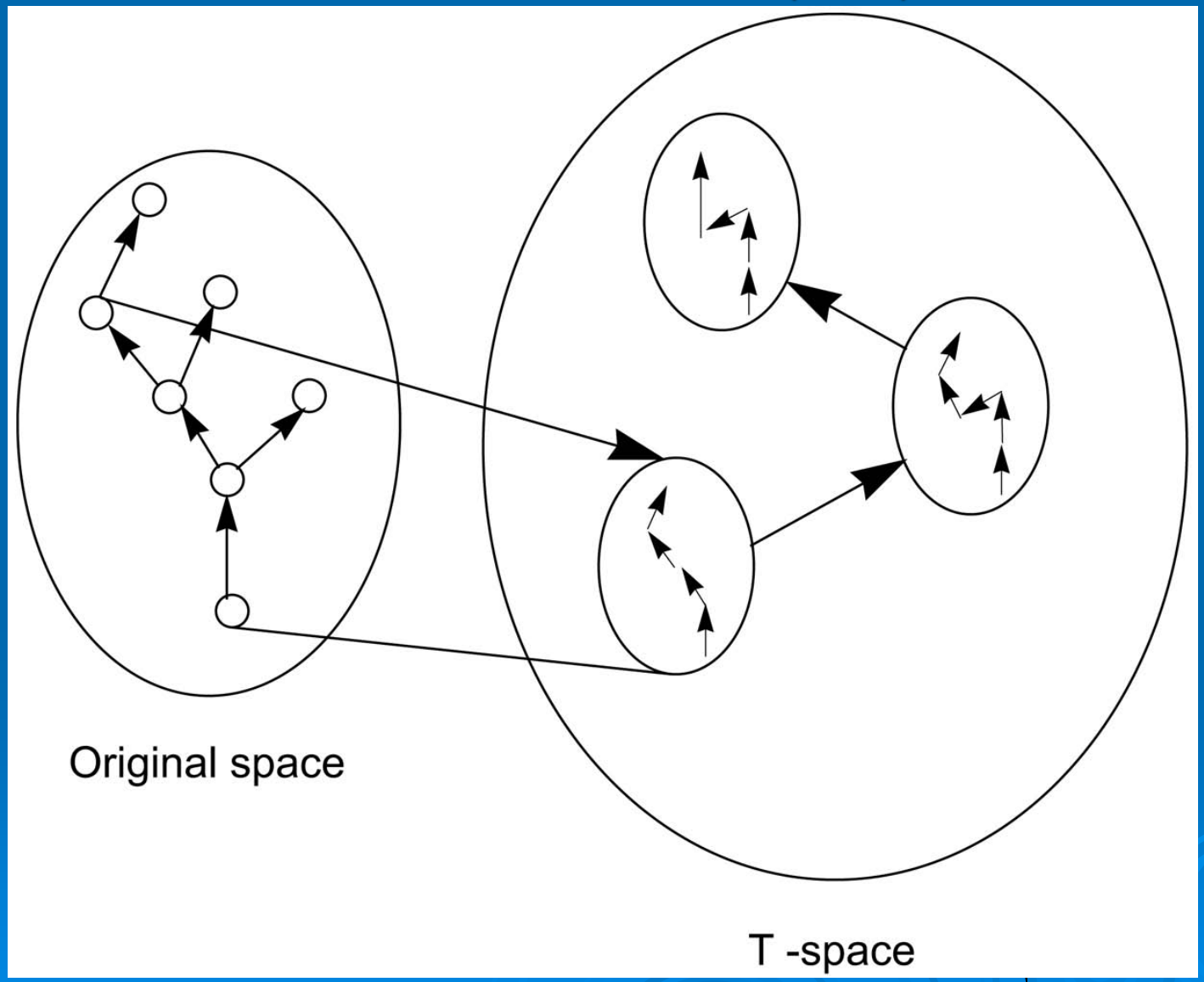
3. *Specify preference.* Look for as exact as possible matches of features before considering more general matches.

4. *Frequency preference.* Check first the most frequently matched cases.

5. *Recency preference.* Prefer cases used most recently.

6. *Ease of adaptation preference.* Use first cases most easily adapted to the current situation.

CBR….Transformational analogy, adapted from Carbonell (1983).



- learning through analogy
- transformational analogy-solve new problem by modifying existing solutions til they may be applied to new instance
- operators modify by inserting, deleting, reordering etc.

The advantages of a rule-based approach include:

1. The ability to use, in a very direct fashion, experiential knowledge acquired from human experts. This is particularly important in domains that rely heavily on heuristics to manage complexity and/or missing information.

2. Rules map into state space search. Explanation facilities support debugging.

3. The separation of knowledge from control simplifies development of expert systems by enabling an iterative development process where the engineer acquires, implements, and tests individual rules.

4. Good performance is possible in limited domains. Because of the large amounts of knowledge required for intelligent problem solving, expert systems are limited to narrow domains. However, there are many domains where design of an appropriate system has proven extremely useful.

5. Good explanation facilities. Although the basic rule-based framework supports flexible, problem-specific explanations, it must be mentioned that the ultimate quality of these explanations depends upon the structure and content of the rules. Explanation facilities differ widely between data- and goal-driven systems.

Disadvantages of rule-based reasoning include:

1. Often the rules obtained from human experts are highly heuristic in nature, and do not capture, functional or model-based knowledge of the domain.
2. Heuristic rules tend to be "brittle" and cannot handle missing information or unexpected data values.
3. Another aspect of the brittleness of rules is a tendency to degrade rapidly near the "edges" of the domain knowledge. Unlike humans, rule-based systems are usually unable to fall back on first principles of reasoning when confronted with novel problems.
4. Explanations function at the descriptive level only, omitting theoretical explanations. This follows from the fact that heuristic rules gain much of their power by directly associating problem symptoms with solutions, without requiring (or enabling) deeper reasoning.
5. The knowledge tends to be very task dependent. Formalized domain knowledge tends to be very specific in its applicability. Currently, knowledge representation languages do not approach human flexibility.

The advantages of case-based reasoning include:

1. The ability to encode historical knowledge directly. In many domains, cases can be obtained from existing case histories, repair logs, or other sources, eliminating the need for intensive knowledge acquisition with a human expert.

2. Allows shortcuts in reasoning. If an appropriate case can be found, new problems can often be solved in much less time than it would take to generate a solution from rules or models.

3. It allows a system to avoid past errors and exploit past successes. CBR provides a model of learning that is both theoretically interesting and practical enough to apply to complex problems.

4. Extensive analysis of domain knowledge is not required. Unlike a rule-based system, where the knowledge engineer must anticipate rule interactions, CBR allows a simple additive model for knowledge acquisition. This requires an appropriate representation for cases, a useful retrieval index, and a case adaptation strategy.

5. Appropriate indexing strategies add insight and problem-solving power. The ability to distinguish differences in target problems and select an appropriate case is an important source of a case-based reasoner's power; often, indexing algorithms can provide this functionality automatically.

The disadvantages of case-based reasoning include:

1. Cases do not often include deeper knowledge of the domain. This handicaps explanation facilities, and in many situations it allows the possibility that cases may be misapplied, leading to wrong or poor quality advice.

2. A large case base can suffer problems from store/compute trade-offs.

3. It is difficult to determine good criteria for indexing and matching cases. Currently, retrieval vocabularies and similarity matching algorithms must be carefully hand crafted; this can offset many of the advantages CBR offers for knowledge acquisition.

Advantages of model-based reasoning:

1. Ability to use functional/structural knowledge of the domain. Increase reasoner's ability to handle a variety of problems, including those that may not have been anticipated by system's designers

2. Model-based reasoners tend to be very robust. For the same reason that humans often retreat to first principles when confronted with a novel problem, model-based reasoners tend to be thorough and flexible problem solvers

3. Some knowledge is transferable between tasks. Model-based reasoners are often built scientific, theoretical knowledge. Because science strives for generally applicable theories, this generality often extends to model-based reasoners

4. Often, model-based reasoners can provide causal explanations. These can convey a deeper understanding of the fault to human users, and can also play an important tutorial role.

The disadvantages of model-based reasoning include:

1. A lack of experiential (descriptive) knowledge of the domain. The heuristic methods used by rule-based approaches reflect a valuable class of expertise.

2. It requires an explicit domain model. Many domains, such as the diagnosis of failures in electronic circuits, have a strong scientific basis that supports model-based approaches. However, many domains, such as some medical specialties, most design problems, or many financial applications, lack a well-defined scientific theory. Model-based approaches cannot be used in such cases.

3. High complexity. Model-based reasoning generally operates at a level of detail that leads to significant complexity; this is, after all, one of the main reasons human experts develop heuristics in the first place.

4. Exceptional situations. Unusual circumstances, for example, bridging faults or the interaction of multiple failures in electronic components, can alter the functionality of a system in ways difficult to predict a priori.

For example, the combination of rule-based and case-based systems can:

1. Offer a natural first check against known cases before undertaking rule-based reasoning and the associated search costs.
2. Provide a record of examples and exceptions to solutions through retention in the case base.
3. Record search-based results as cases for future use. By saving appropriate cases, a reasoner can avoid duplicating costly search.